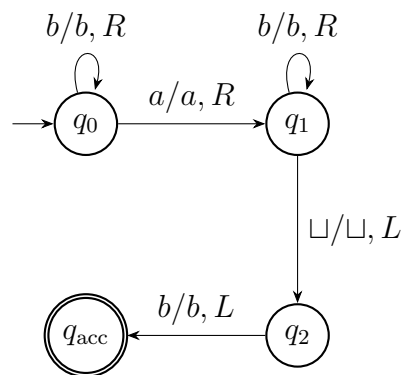


## Quiz 2 — April 28, 2026

CPSC 406 — Algorithm Analysis,  
Chapman University · Spring 2026  
Instructor: Jonathan Weinberger

Name: \_\_\_\_\_

1. (2 points) (a) Consider the following Turing machine over input alphabet  $\Sigma = \{a, b\}$ . The blank symbol is  $\sqcup$ . All omitted transitions go to a rejecting halt state.



Which of the following words is accepted by the machine? Tick exactly one.

- ☐  $abb$   
☐  $aab$   
☐  $aabb$   
☐ All of the above.  
☐ None of the above.

(b) Let

$$f(n) = 4n^3 + 8n \log n + 1.$$

Which of the following Big-O classes contain  $f(n)$ ? Tick exactly one.

- ☐  $\mathcal{O}(n^3)$   
☐  $\mathcal{O}(n^3 \log n)$   
☐  $\mathcal{O}(\exp(n))$   
☐ All of the above.  
☐ None of the above.

2. (2 points) Show that the following formula, given in CNF, is unsatisfiable using resolution:

$$\varphi = (p \vee q) \wedge (\neg p \vee r \vee s) \wedge (\neg q \vee r) \wedge (\neg r \vee t) \wedge (\neg s \vee t) \wedge \neg t \wedge \neg q.$$